Chat Transcript

2026-11-11 — Recursion vs. DFS

User

For this task which option would be more efficient? Recursion or DFS?

Assistant

For this task, use **iterative DFS**.

Recursion and DFS are not mutually exclusive—recursive traversal is also DFS. Both have:

- **Time:** `O(V + E)`
- **Space:** `O(V)`

However, iterative DFS is more efficient and safer in Java because it:

- Avoids stack overflow for deep hierarchies
- Makes the `visited` set explicit for preventing duplicates/cycles
- Avoids recursive call overhead

Use a `Deque<Employee>` as a stack and mark employees as visited when adding them, not when removing them. This matches the requirements in `Task1.md`.

2026-11-11 — DFS Implementation

User

Create the DFS implementation of this solution as described in `Task1.md`

Assistant

Implemented DFS reporting structure support.

Changes

- Added `ReportingStructure` DTO with `employee` and `numberOfReports`.
- Implemented iterative DFS with:
  - `Deque<Employee>` stack
  - Visited employee IDs to prevent duplicates/cycles
  - Repository resolution for nested report references
- Added integration test validating John Lennon has 4 reports.
- Documented endpoint in README.

Files changed:

- `src/main/java/com/mindex/challenge/data/ReportingStructure.java`
- `src/main/java/com/mindex/challenge/service/impl/EmployeeServiceImpl.java`

- `src/test/java/com/mindex/challenge/service/impl/EmployeeServiceImplTest.java`
- `README.md`

Validation passed:

```
BUILD SUCCESSFUL
```

2026-11-11 — Transcript Export

User

Reconstruct all of my chats and your responses into an md file

Assistant

Created this Markdown transcript containing the user and assistant messages available in the current conversation.